were having a first meeting on this. Give me points to note and things to discuss on how to implement this project based on Ghana's LI 2404 and the Mincom Ghana ExcaVATORE tracking project

Dear Mrs N'GONIAN,
Good morning, and thank you for your message. My name is Dr Sylvester Akpah, the consultant for the Minerals Commission Small-Scale Monitoring System.
I am delighted that your delegation found the visit to the Minerals Commission of Ghana worthwhile, and I welcome the interest of Côte d'Ivoire in a similar system for monitoring small-scale mining.
Smart Innovations Ghana Limited designed and operates the monitoring system you

Show more

Preparing meeting points on Ghana's LI 2404 tracking project.

I'll check what LI 2404 and ExcaVATORE actually cover so the meeting points rest on the real rules.

Searched the web

Below are briefing notes for the first meeting with Côte d'Ivoire's mining directorate. The ExcaVATORE name didn't turn up in public sources, so I've treated it as the Ghana equipment-tracking programme under LI 2404. The drone surveillance component in Dr Akpah's email also isn't covered in what I found, so I've left it out of the Ghana facts.

What Ghana's model rests on

- **Legal basis:** LI 2404 is the Minerals and Mining (Mineral Operations, Tracking of Earth-Moving and Mining Equipment) Regulations, 2020. It requires all mining-related earth-moving equipment to be registered and operated only within licensed mining concessions. modernghana
- **Registration:** Owners must state the purpose and area of use, and unregistered excavators are liable to seizure and confiscation. District assemblies recorded the owner's details, entry date, registration and chassis numbers, and the operator's details. ghanaiantimes ghanaiantimes
- **Control room:** The 24/7 Network Control Centre at the Minerals Commission also tracks explosive-carrying vehicles and runs a call centre for complaints. gna
- **Geo-fencing logic:** The software pairs each excavator with one licensed small-scale concession at a time, with a 15 metre buffer around it. Concession site-plan coordinates were loaded into the Ghana Mine Repository and Tracking software. graphiconline graphiconline
- **Field marking:** Once a tracker is fitted, a sticker with the control room's phone number goes on the machine. graphiconline
- **Import control:** Permits for imported equipment are granted only after the Minerals Commission and DVLA verify the intended end-use. DVLA and the Commission run a 24-hour office at Tema Port. graphic graphic
- **Scale-up:** The control centre was set up in December 2022, and by early 2023 it was monitoring 75 excavators. By the latest update, 4,300 units were registered at Tema Port, with trackers on 1,864. peacefmonline modernghana
- **Inter-agency set-up:** Lands, Transport, GRA, DVLA and the Minerals Commission carry out the programme together. modernghana

Points to note

1. **The law came first.** The tracking system enforces a regulation that makes registration and in-concession operation mandatory. Without an equivalent legal basis in Côte d'Ivoire, a tracker is only a monitoring tool with no enforcement power. This is the biggest dependency in the project.
2. **It is more than a technology project.** Ghana's results depended on customs, port, vehicle licensing, local government and security agencies all acting together. The control room only flags; others have to act on the flag.
3. **Concession data quality drives everything.** Geo-fencing works only if licence boundaries are digitised and accurate. Ghana had to load site-plan coordinates for every legal concession.
4. **Ghana started small.** It went from about 40 to 75 machines in the pilot phase to thousands later. This supports your pilot-first approach.
5. **Enforcement is the weak point.** More than 3,000 excavators registered under the programme remain unclaimed at Tema Port after geo-fencing was introduced, which suggests some owners may be avoiding a regime that restricts where machines can work. Expect Côte d'Ivoire to ask about this, so have an honest answer ready. graphic
6. **Sustainability was flagged early.** The Lands Minister asked the Commission for a sustainability plan for the initiative. Cost-recovery and funding of device, airtime and control-room costs should be in your proposal from the start. graphic
7. **Be careful with claims.** Don't promise "real-time flagging" or drone coverage figures unless you can back them with Ghana performance data. The delegation will compare notes with what they saw in Accra.

Things to discuss

Legal and institutional

- Which Ivorian law governs small-scale and semi-industrial mining, and does it cover equipment registration and tracking? If not, who would draft the regulatory instrument, and by when?
- Which body would own the system and the control room, and how would it work with other agencies?
- Who would handle equipment imports and customs, and could an import-permit step be

added as in Ghana?

Data and operations

- What is the state of licence and permit data (digitised, GPS-referenced, complete)?
- How many machines are involved, how are they distributed, and which zones are the priority, including border and forest areas?
- Which agencies would respond to alerts (police, gendarmerie, environment, water and forests), and what is the protocol after an alert?
- How would the system cover machines already in the field, and how would tagging teams be protected?

Technical

- What mobile coverage and power exist in the mining zones, and what device specification does that require?
- Data hosting and sovereignty: should data sit in Côte d'Ivoire?
- System interoperability with existing mining cadastre or licensing databases.
- Language and interface: French-language system and training.

Commercial and delivery

- Who funds the needs assessment, the pilot and the rollout?
- Procurement route (direct award, tender, or partnership) and expected timeline.
- Who owns the system and the data at the end, and what training and handover are planned?
- Pilot scope: which regions, how many machines, how long, and what success metrics?

Suggested closing for the meeting

Agree on a point of contact at the directorate, the list of data and documents they will share (legal texts, licence registers, equipment estimates), a date for the needs-assessment kickoff, and a shortlist of two or three candidate pilot regions.

If it would help, I can turn this into a one-page agenda or a French-language version for sharing with Mrs N'Gonian's team.

Sources:

- Over 3,000 excavators remain unclaimed at Tema Port (Graphic Online)
- Govt moves to track excavator imports (Graphic Online)
- Mining equipment to be fitted with tamper-proof GPS tracker (Graphic Online)
- MMDAs must register all excavators (Ghanaian Times)
- Lands Minister inaugurates Network Control Centre (GNA)
- Excavator tracking centre launched (Graphic Online)
- Tracking of excavators (Peace FM)

25 minutes ago

what'll Cote Dviore need to implement same

Outlining what Côte d'Ivoire would need to adopt Ghana's model.

Based on how Ghana built it, Côte d'Ivoire would need these things. I'm confident of the Ghana side; for Ivorian specifics I've marked what to confirm in the meeting.

1. Legal and policy foundation

- **A binding rule on equipment.** This would be a decree, regulation or ministerial order requiring registration of earth-moving equipment used in mining, restricting it to licensed areas, and setting penalties such as seizure. Ghana's LI 2404 does this.
- **Legal power for the regulator** to fit trackers, access equipment, and act on violations. Confirm whether the current mining code (and its small-scale or semi-mechanised permit categories) already gives the directorate this authority or needs a new instrument.
- **Rules on data and privacy** for GPS tracking of private owners' machines, and on evidence admissibility for alerts.

2. Institutional set-up

- **A lead agency and a mandate**, probably the mining directorate, with a named project owner.
- **Inter-agency agreements** with customs, the transport and vehicle registration authority, local government, the ports authority and security forces. Ghana's system works because these agencies act on what the control room finds.
- **A response protocol:** who is called, within what time, and with what powers when an alert fires.

3. Data

- **A digitised, GPS-referenced register of all valid licences and concessions.** This is the base layer for geo-fencing.
- **An inventory of existing equipment** (owners, types, chassis and registration numbers) and a way to capture new imports at the border and port.
- **Clear definitions of permitted areas**, including buffers, forest reserves and no-go zones.

4. Technology

- **Tamper-resistant GPS trackers** suited to local conditions (vibration, dust, heat, battery life).
- **A tracking and geo-fencing platform** that pairs each machine to its licence and generates alerts.
- **A control room** with 24/7 staffing, hardware, power backup, connectivity, and a hotline or call centre.

- **Data hosting and security**, ideally in-country, with backups and user access controls.
- **Connectivity** (mobile or satellite) in remote mining zones, which needs checking before pilot sites are chosen.
- **Drones and AI imagery**, if included, add their own needs: pilots, flight permissions, image processing and a way to link findings to enforcement.

5. People and capacity

- Control room operators and supervisors.
- Field teams to install and inspect trackers, with security escort in risky zones.
- Trainers and a French-language curriculum, plus a handover plan so the directorate can run the system without the vendor.

6. Money

- **Upfront costs:** devices, installation, platform set-up, control room build-out.
- **Running costs:** airtime or data, device replacement, staff, maintenance, licensing.
- **A funding model:** state budget, donor or climate-finance support, or a fee charged to equipment owners. Ghana's minister explicitly called for a sustainability plan, so decide this early.

7. Delivery plan

- Needs assessment, then a pilot in selected regions, then phased national rollout, with agreed success measures at each stage (machines registered, share of alerts acted on, violations reduced).
- Communication to miners and equipment dealers before enforcement starts, since compliance depends on people understanding the rules.

What to confirm with them first

1. Whether the current legal framework is enough or a new regulation is required, and how long that would take.
2. Whether licence data is digitised and accurate.
3. Which agencies they can get to commit.
4. Who pays.

If the legal basis and funding are unresolved, you can still do the needs assessment, but the pilot will be hard to launch.

23 minutes ago

WHat are the benefits to theme if they are to implement same

Explaining economic benefits for Côte d'Ivoire's exports.

These are the benefits Côte d'Ivoire could reasonably expect, and I've separated what Ghana's experience supports from what is still a projection.

Benefits

1. Enforcement against illegal mining

- Machines can be tied to a specific licence and flagged when they leave it, so the authority no longer depends on patrols and tip-offs alone.
- Registration at import and the permit step make it harder to bring in equipment with no legitimate use. Ghana's approach checks the intended end-use before equipment is released.
- The sticker and control-room number gives communities and officials a visible way to tell a legal machine from an untracked one and to report it.

2. Environmental protection

- Less damage to rivers, forests and farmland from machines operating where they shouldn't.
- Faster detection of new illegal sites, especially if drone imagery is added.
- A clearer link between a machine, its owner and the damage it causes, which supports holding owners responsible for reclamation. Ghana's Minerals Commission cited this accountability goal when it launched the programme.

3. Revenue and formalisation

- A reliable register of mining equipment and where it works gives the state a clearer picture of the sector, which helps with licensing, royalties and tax collection.
- Operators have a stronger incentive to get licensed, since an unlicensed machine can't work freely.
- Ghana's minister linked the sharp rise in small-scale gold exports (from about 3,430 kg in 2021 to about 22,158 kg in 2022) to a combination of measures, including a tax cut and tracking. That is not evidence that tracking alone caused it, so I'd avoid presenting it to them as a proven result.

4. Better data for decision-making

- One system holding licences, equipment and movements allows planning of priority zones and resource allocation.
- Records of equipment activity help in disputes, prosecutions and licence renewals.

5. Security and safety

- Tracking can extend to other high-risk items. Ghana's control centre also monitors explosive-carrying vehicles, which matters for safety after incidents such as the Appiatse explosion.
- Fewer confrontations in the field, because enforcement is targeted rather than based on

broad patrols.

6. Institutional and political gains

- Shows a visible, technology-based response to illegal mining, which supports public and donor confidence.
- Builds local expertise in the directorate through training and handover.
- Opens the way to regional cooperation, since mining equipment and illegal mining often cross borders.

Caveats to be honest about

- The benefits depend on enforcement. Ghana still has thousands of excavators unclaimed at the port, which shows owners may avoid the regime and that tracking alone doesn't solve the problem.
- Results build over time. Ghana moved from tens of machines to thousands over a few years.
- The strongest case is a combined one: legal backing, agency cooperation, data quality and funding together, rather than the technology on its own.

For the meeting, I'd frame the benefits as targets to be measured in the pilot, for example the number of machines registered, the share of alerts acted on, and the area of illegal activity detected, rather than as guaranteed outcomes.

Create a doc

Reply

Claude is AI and can make mistakes. Sonnet 5.5 Medium Manual